

MITCHELL, SD, USA

WMO: 726545

Lat: 43.774N

Lon: 98.038W

Elev: 396

StdP: 96.66

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-20.9	-28.1	0.3	-22.7	-25.0	0.4	-20.2	15.6	-11.8	13.9	-7.6	4.4	330	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	34.3	23.4	32.6	22.8	31.1	22.1	25.3	31.5	24.3	30.3	23.4	29.2	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	19.3	29.0	22.5	18.0	28.1	21.4	16.9	27.0	79.9	31.2	75.6	30.1	71.6	29.0	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.7	11.2	9.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-27.4	37.6	3.9	2.5	-30.2	39.4	-32.5	40.9	-34.6	42.4	-37.5	44.2		
			-27.2	27.0	3.3	1.3	-29.5	27.9	-31.5	28.7	-33.3	29.4	-35.7	30.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-7.5	-5.2	1.2	8.4	15.1	20.9	24.0	22.2	17.9	10.1	1.9	-5.3
	DBStd	12.36	7.37	7.39	7.30	5.65	4.85	3.86	3.53	3.14	4.77	5.34	6.23	6.65
	HDD10.0	2144	542	425	282	94	11	0	0	0	3	65	250	474
	HDD18.3	3988	800	658	530	299	124	18	3	5	66	260	493	732
	CDD10.0	1681	0	0	11	47	169	328	435	379	239	68	6	0
	CDD18.3	484	0	0	0	2	24	96	180	126	52	4	0	0
	CDH23.3	5059	0	0	7	56	290	964	1941	1172	558	69	2	0
	CDH26.7	1983	0	0	1	13	91	354	876	429	202	19	0	0
(14) Wind	WSAvg	4.7	4.9	5.0	5.2	5.5	5.1	4.4	4.0	3.9	4.3	4.8	4.8	4.8
(15) Precipitation	PrecAvg	556	11	15	42	64	81	93	63	57	61	38	24	16
	PrecMax	828	39	74	190	180	255	282	134	168	174	117	95	50
	PrecMin	329	0	0	4	9	18	10	7	7	3	0	0	0
	PrecStd	131	10	17	37	39	56	57	35	34	37	32	22	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	15.8	23.9	28.2	32.7	35.1	38.5	35.8	33.9	29.1	21.9	12.6
		MCWB	6.2	8.8	13.1	16.1	20.1	23.6	23.8	24.0	21.4	17.1	12.3	6.3
	2%	DB	7.0	11.0	19.1	24.4	28.9	32.4	35.4	32.7	31.5	24.9	17.6	8.5
		MCWB	3.2	6.1	11.1	14.1	18.6	22.3	24.0	23.5	20.9	15.6	10.4	4.2
	5%	DB	4.1	7.4	15.2	21.3	26.7	30.6	33.0	30.9	28.9	22.0	13.7	5.4
		MCWB	1.7	3.9	9.3	12.4	17.3	21.5	23.8	22.5	19.9	14.2	8.3	2.1
	10%	DB	2.4	4.4	12.0	18.2	23.9	28.6	31.3	28.9	26.8	18.8	11.0	2.9
		MCWB	0.6	1.9	7.3	10.9	16.0	20.3	22.9	21.7	18.8	12.5	6.3	0.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	9.8	15.0	17.5	21.8	25.5	26.9	26.2	23.7	19.2	13.6	7.4
		MCDB	11.8	15.7	19.7	26.5	29.6	32.4	33.1	32.3	29.7	25.8	19.4	11.6
	2%	WB	3.5	6.1	12.3	15.2	20.0	23.9	25.7	24.9	22.5	17.0	10.9	4.3
		MCDB	6.3	10.1	17.9	22.7	26.8	30.5	31.9	30.5	28.9	22.4	16.4	8.1
	5%	WB	1.8	4.0	10.0	13.2	18.6	22.6	24.8	23.9	21.3	15.0	8.6	2.5
		MCDB	3.9	7.3	15.5	19.8	25.2	28.6	31.0	29.2	27.4	20.8	13.7	5.2
	10%	WB	0.6	2.0	7.2	11.6	17.0	21.4	23.8	22.8	20.0	13.0	6.5	1.0
		MCDB	2.3	4.4	11.5	17.3	22.6	27.1	29.9	27.6	25.1	18.1	10.7	2.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.3	10.9	12.1	11.8	11.5	12.2	11.9	13.1	12.1	11.1	9.1
		MCDBR	12.2	13.9	17.2	17.9	15.6	13.8	14.2	14.1	16.0	16.9	16.4	12.7
	5% WB	MCWBR	9.3	9.7	10.4	9.3	7.6	6.3	5.8	6.5	7.3	9.0	10.0	9.2
		MCWBR	11.6	13.4	16.7	15.6	13.9	12.3	12.6	12.0	13.5	14.7	16.0	12.2
(40) Clear-Sky Solar Irradiance	taub		0.269	0.287	0.313	0.364	0.384	0.387	0.395	0.399	0.351	0.318	0.288	0.265
		taud	2.449	2.424	2.436	2.301	2.331	2.372	2.358	2.328	2.438	2.499	2.493	2.472
	Ebn at Noon		874	915	930	899	888	883	871	854	876	867	843	846
		Edh at Noon	71	88	99	124	124	119	120	119	99	81	69	64
(44) All-Sky Solar Radiation	RadAvg		1.90	2.92	3.86	4.89	5.74	6.44	6.77	5.63	4.48	3.01	2.05	1.51
	RadStd		0.16	0.21	0.34	0.41	0.45	0.46	0.22	0.43	0.33	0.35	0.18	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page